

FIR FILTER FOR ECG SIGNAL

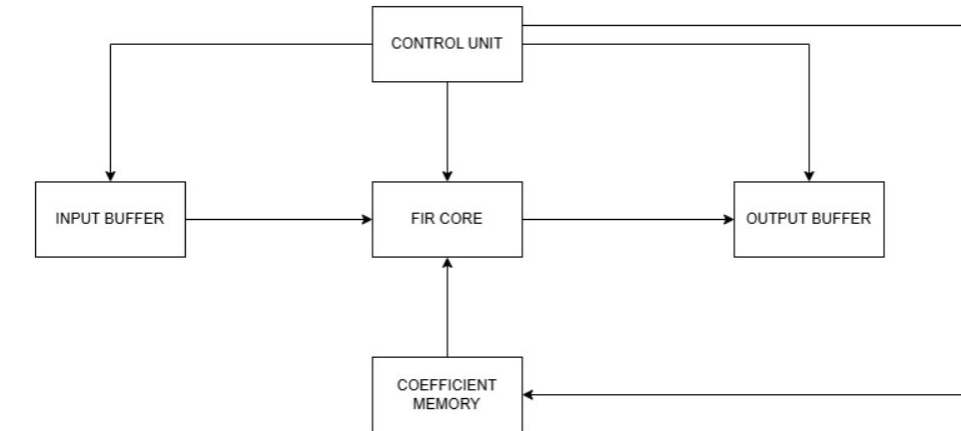
Problem statement

Electrocardiogram (ECG) signals are often affected by unwanted noise and interference during signal acquisition, which can reduce the accuracy of heart monitoring and diagnosis. This project aims to design and implement a Finite Impulse Response (FIR) filter for ECG signal processing using Vivado through FPGA kit.

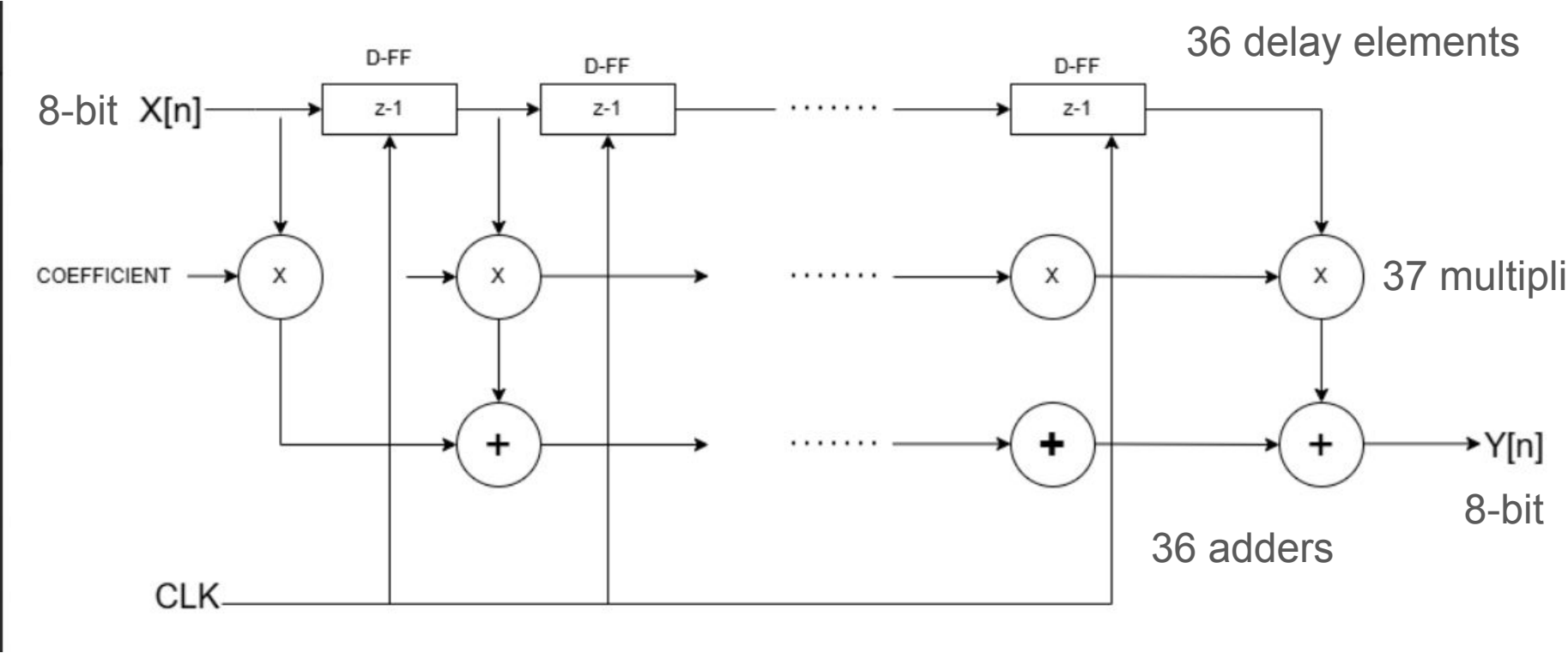
Objectives

- 1)To design an FIR filter for reducing noise in ECG signals.
- 2)To implement the FIR filter using Verilog HDL.
- 3)To simulate and verify the design using Xilinx Vivado.

Block diagram



Architecture



Formula used:


$$y[n] = x[n] ** h[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n - k]$$

Parameters considered:

Sampling frequency= 360Hz

Cutoff Frequency= 30Hz

ORDER OF FILTER N = 36

NUMBER OF TAPS = 37

NUMBER OF DELAY ELEMENTS = 36

NUMBER OF ADDERS = 36

NUMBER OF MULTIPLIERS = 37

INPUT SIZE = 8 BIT

OUTPUT BIT SIZE = 8 BIT

Coefficient=

Github

<https://github.com/rasheedfouzan/FIR-FILTER-FOR-ECG.git>

PBL WORK DISTRIBUTION:

- FIRSTLY A GOOGLE DOCS WAS CREATED TO TRACK THE PROGRESS
- EACH PERSON WAS GIVEN THE TASK OF READING AND ANALYSING A RESEARCH PAPER AND THE INITIAL OUTCOME OF THE RESEARCH BY THE MEMBERS HAVE BEEN UPLOADED
- BASED ON THE DISCUSSIONS THE FOLLOWING PAPER WAS SELECTED AS THE BASE PAPER :**Hardware Implementation of FIR Filter for ECG Signal Processing: Design, Optimization, and Performance Analysis on an FPGA**
- THE LEAF CELLS WERE IDENTIFIED SUBSEQUENTLY AND THEY WERE AS FOLLOWS:DELAY ELEMENT,ADDER,AND MULTIPLIER
- THE TASKS OF IN-DEPTH RESEARCH INTO EACH LEAF CELL WAS DIVIDED AS FOLLOWS:
 - DELAY ELEMENT – FREYA
 - MULTIPLIER – FOUZAN
 - ADDER – DHRUTHWIK
- ALL THE TEAM MEMBERS SUBSEQUENTLY DECIDED AFTER RESEARCH THAT THE IDEAL NUMBER OF TAPS FOR AN FIR FILTER WOULD BE WITHIN THE RANGE OF 35-40 AND HENCE CHOSE 37
- EACH MEMBER THEREAFTER GENERATED THE CODE AND TESTBENCH TO TEST